R2

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1 Questions/Theories/Ponderings

1.1 PN Detection when looking for BH-Sol systems

This was a fun idea, but PNe are most likely wayyy too short-lived (assuming a timescale of 10-25kyr) If I got nothing else to do I'll do the math for it. Speaking of, this is sorta an important problem, what's the method/formula for this? I.e finding the probability that a transient event is visible?

In previous papers they looked solar for tidal deformations from another orbiting BH on the sunlike star. However, wouldn't it also make sense to factor in the fact that said systems *must* have evolved through CE evolution and ejected a shell? Thus producing some sort of PN ring? I'll have to look into it more, as I'd imagine said ring would also be ionized for a brief(?) period of time before then largely acting to obscure (and maybe redden) the sun-like star

1.2 POSYDON Dataset to Classify Tertiary System

Couldn't one theoretically use a large POSYDON population and its properties to then compare to the properties of observed systems to determine if they evolved through binary evolution? Accidentally stumbled across this around a year ago with V404 Cygni, but it feels relatively feasible to make a grouping algorithm to find systems which are predicted to have evolved through tertiary means (or capture/dynamic methods)

1.3 Striation in Mass Loss Graphs

In the graphs with ΔM_{SNe} we see some striation happening with masses loses between 3.5 to 3.7 and then again between 3.9 to 4.1, is this just a byproduct of POSYDON resolution or is there some underlying physics concept?

1.4 Resolution (?) Woes

In the BH-Sol Masses at BH Formation graph 2 we can see the effects of the resolution of the POSYDON reference grids (could also possibly be some sort of underlying physics), also, notably, there doesn't seem to be a massive difference between the lower resolution one ($.0Z_{\odot}$) and the higher res.

1.5 POST CE Eccentricity

Sorta can? By process of elimination it's because of the SNe, but, the natal kick velocity doesn't seem to have a strong correlation, yet the systems y_vel does?

Cannot for the life of me figure out what is causing the distribution of eccentricities for the BH-Sol systems. It seems to vaguely be a normal distribution, but (so far) I haven't found any properties that seem to be correlated. The CE stage sets eccentricity to 0 at envelope ejection, which really only leaves the SNe stage, but the mass loss and the natal kick velocity don't seem to be correlated.

2 10 Mill Pop Graphs

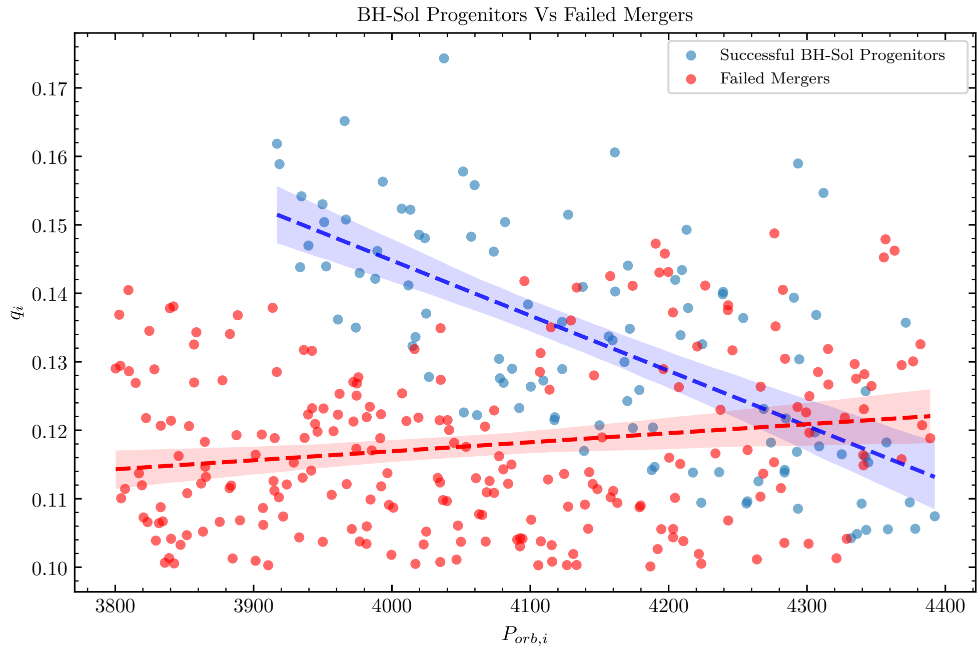
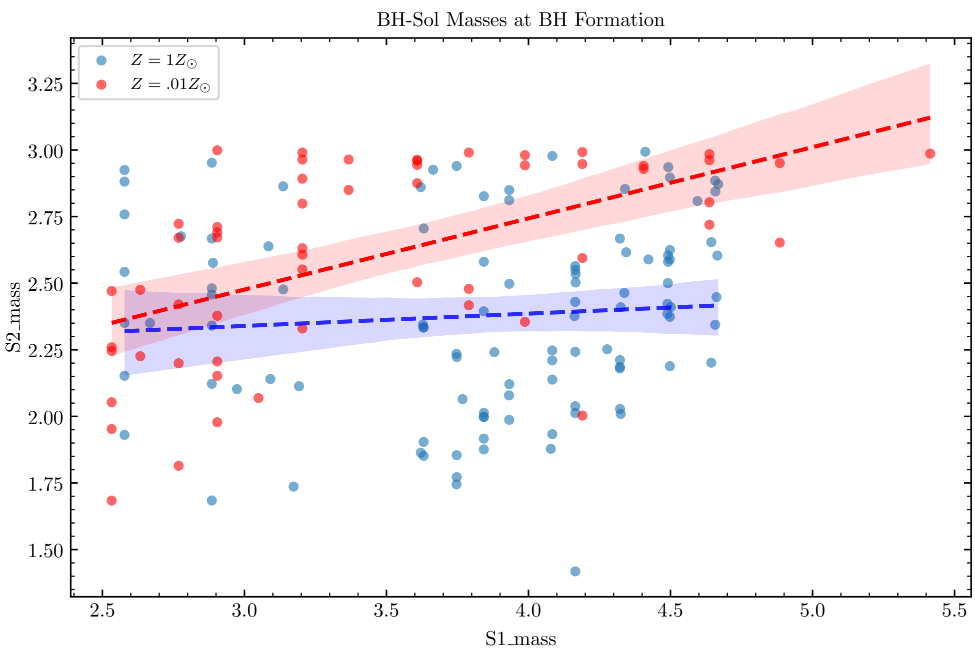


Figure 1: This shows a clear distinction between the failed vs survived BH-Sol progenitors.



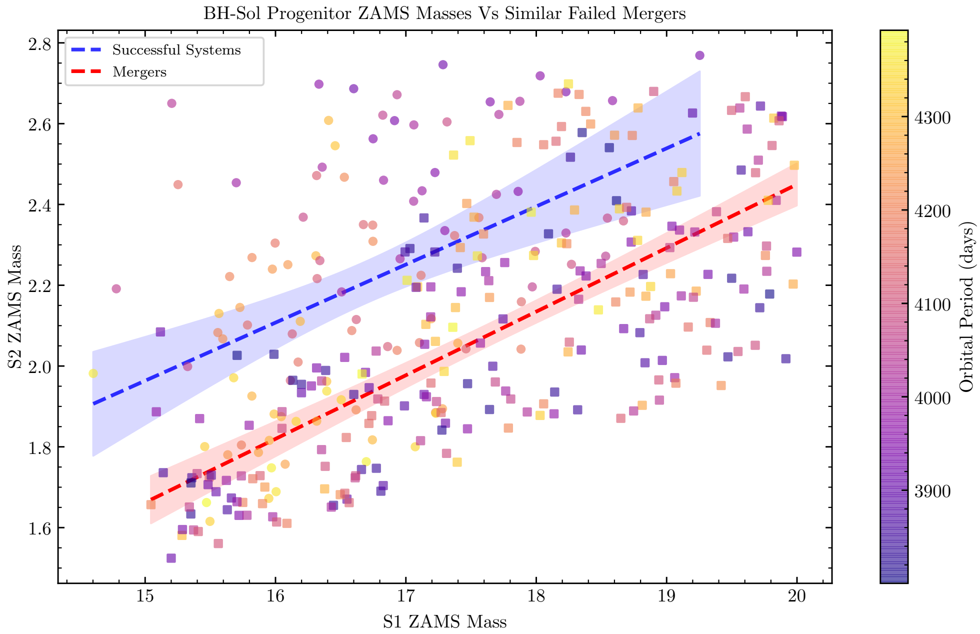
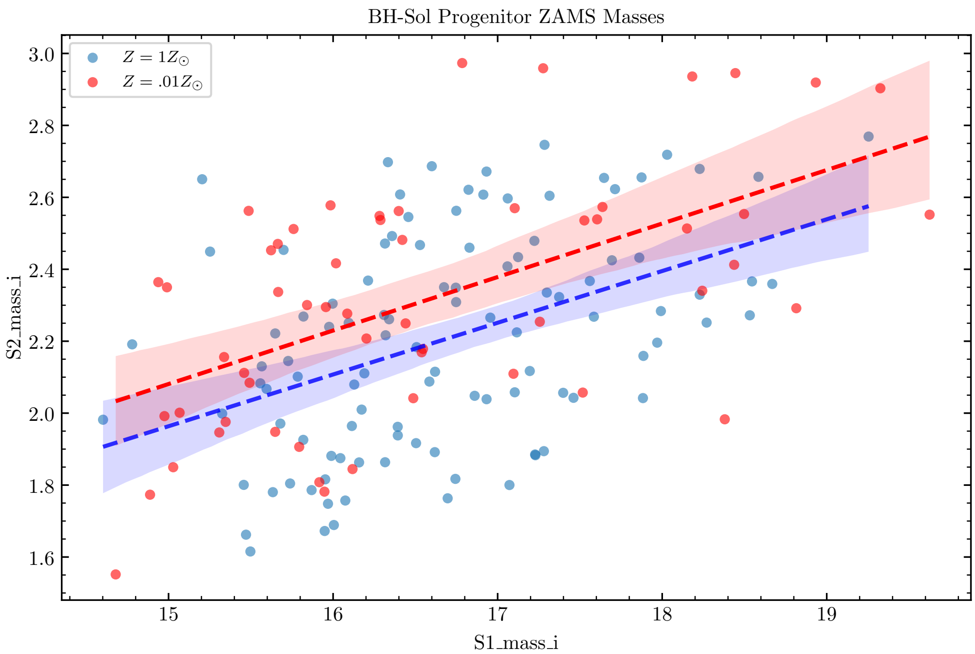
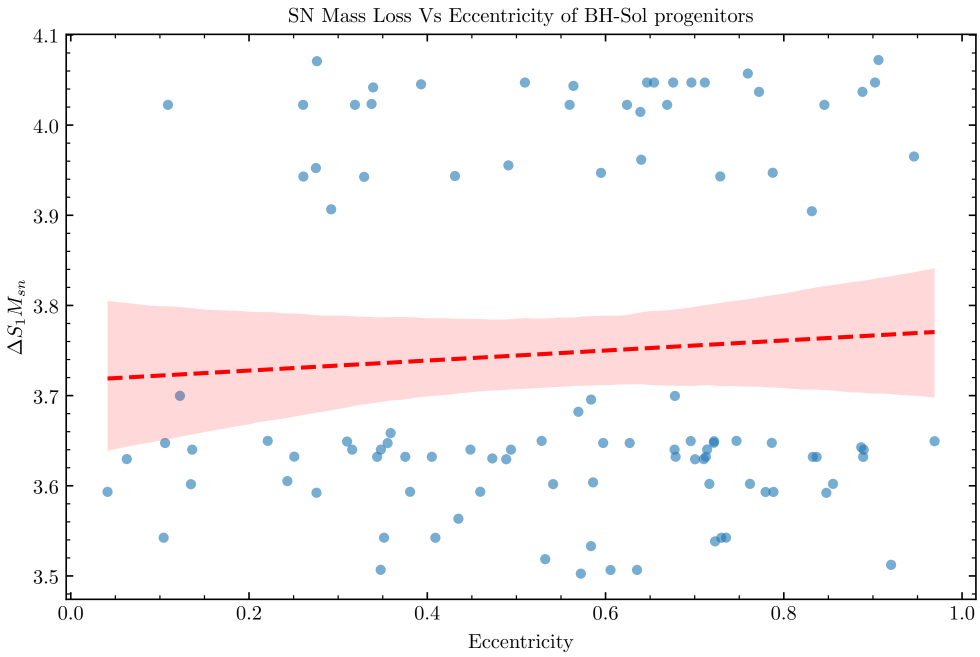
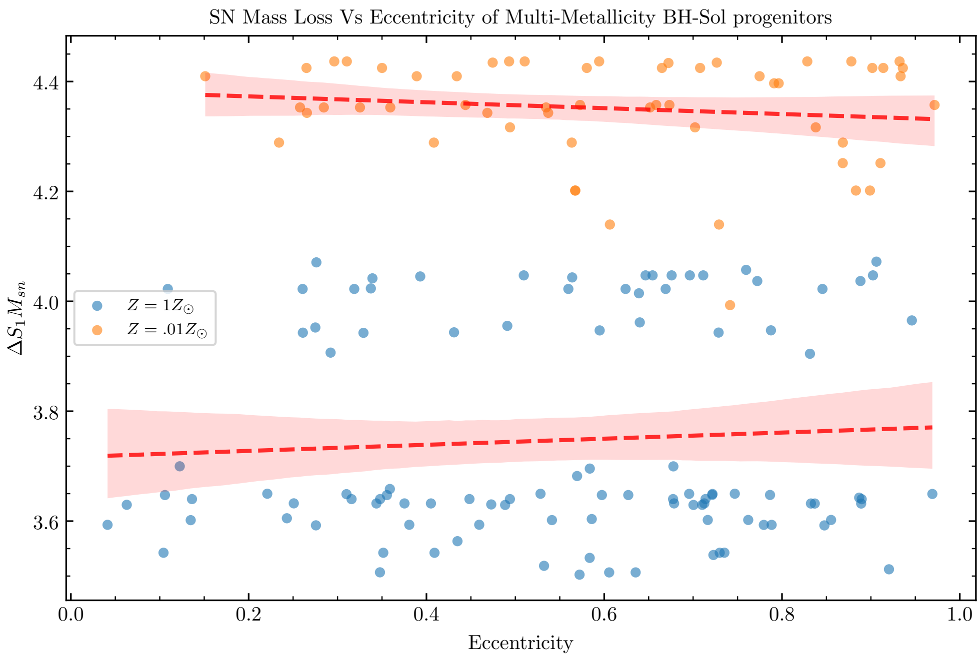
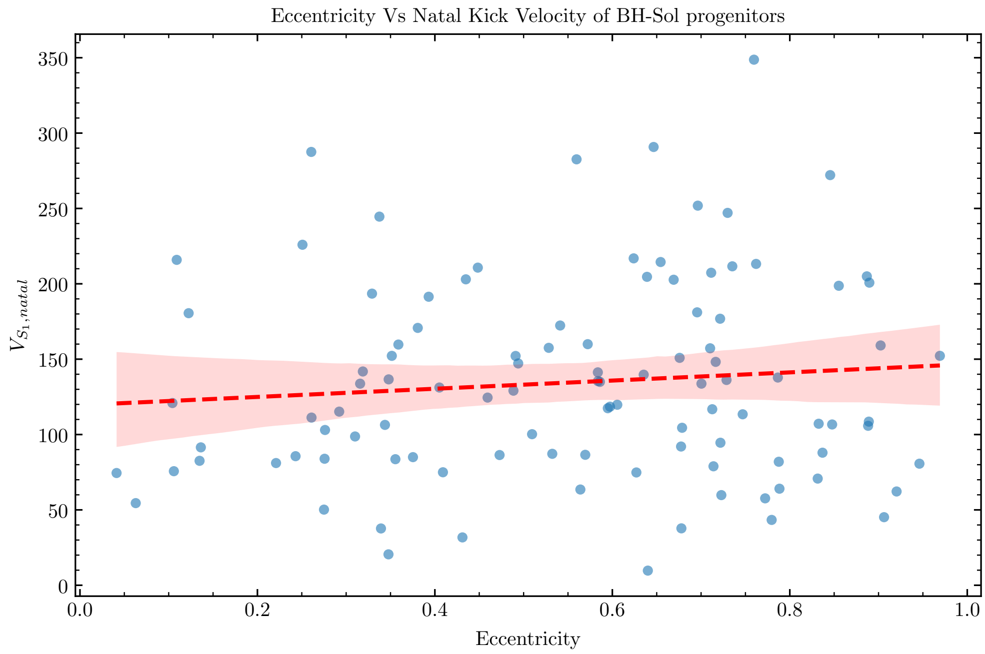


Figure 2: Here we can see that there seems to be a (quite rough) survival line based on the initial mass ratio and the orbital period. Note that these were simply constrained based on the max and min of the initial masses. Next step is actually sampling from a proper matching distribution.

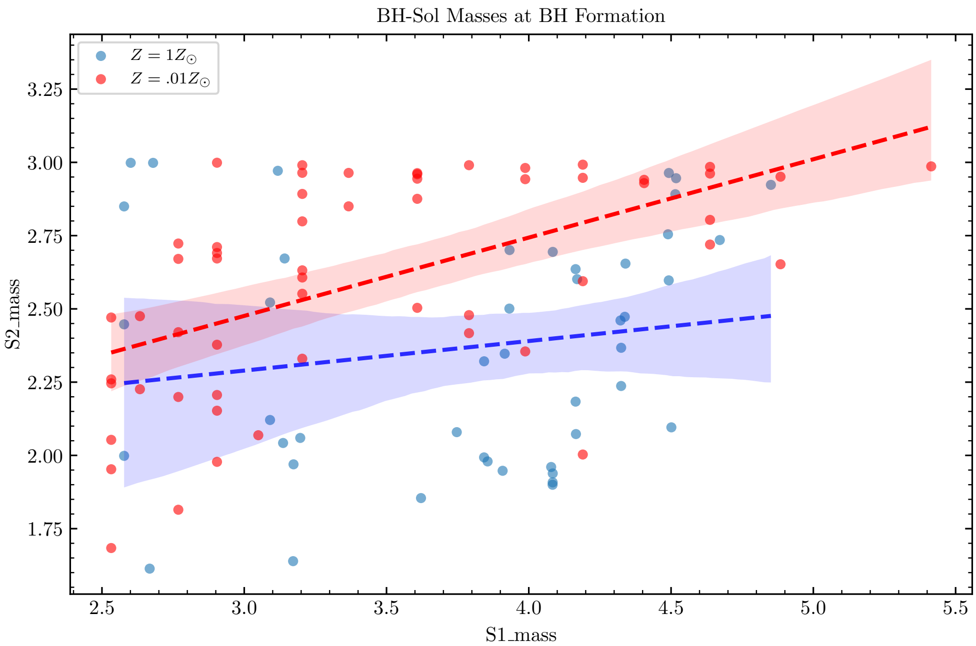


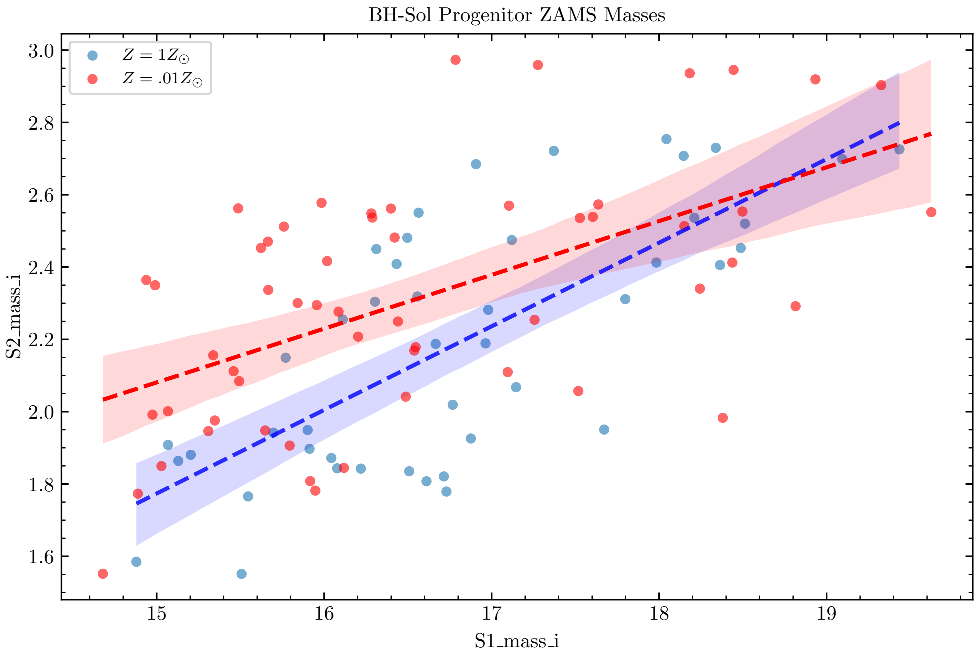

SN Mass Loss Vs Eccentricity of BH-Sol progenitors
$\Delta S_1 M_{sn}$
4.1
4.0
3.9
3.8
3.7
3.6
3.5
0.0
0.2
0.4
0.6
0.8
1.0
Eccentricity

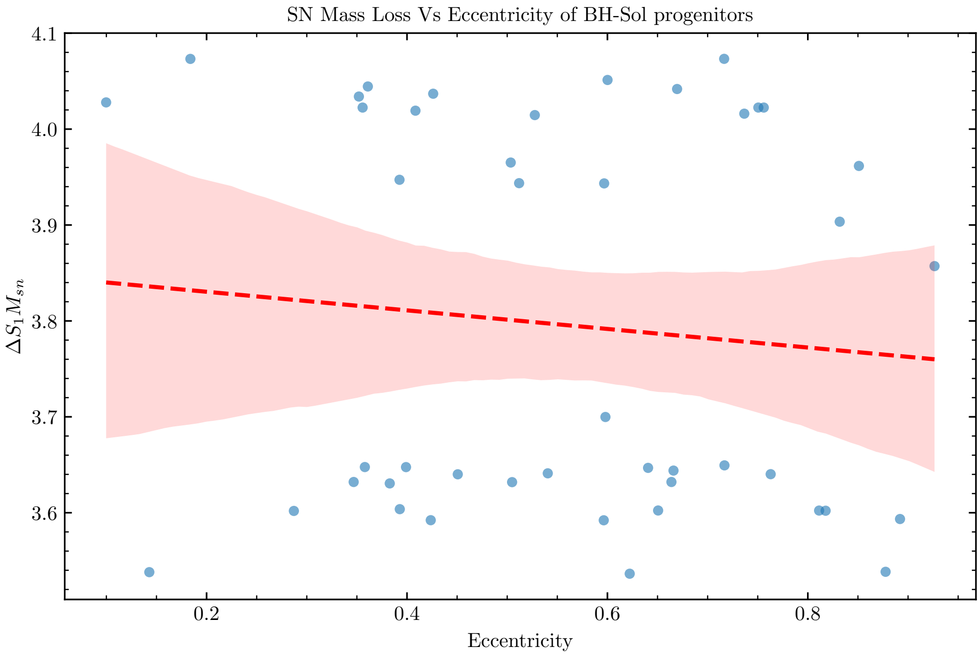

SN Mass Loss Vs Eccentricity of Multi-Metallicity BH-Sol progenitors
$\Delta S_1 M_{sn}$
4.4
4.2
4.0
3.8
3.6
$Z = 1Z_\odot$
$Z = .01Z_\odot$
0.0
0.2
0.4
0.6
0.8
1.0
Eccentricity


Eccentricity Vs Natal Kick Velocity of BH-Sol progenitors
$V_{S_1,natal}$
0
50
100
150
200
250
300
350
0.0
0.2
0.4
0.6
0.8
1.0
Eccentricity

3 1 Mill Graphs



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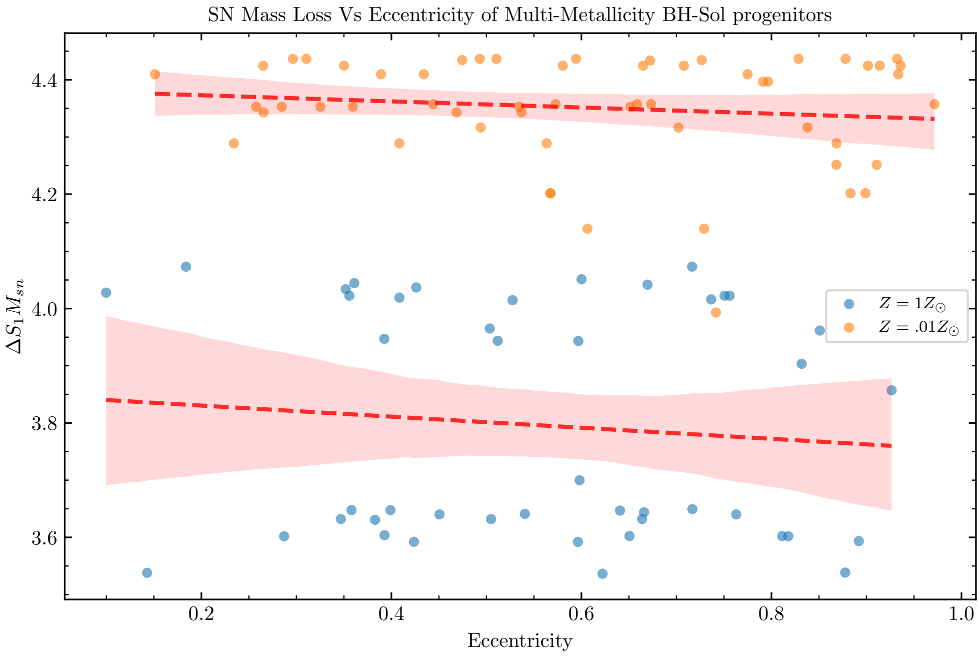
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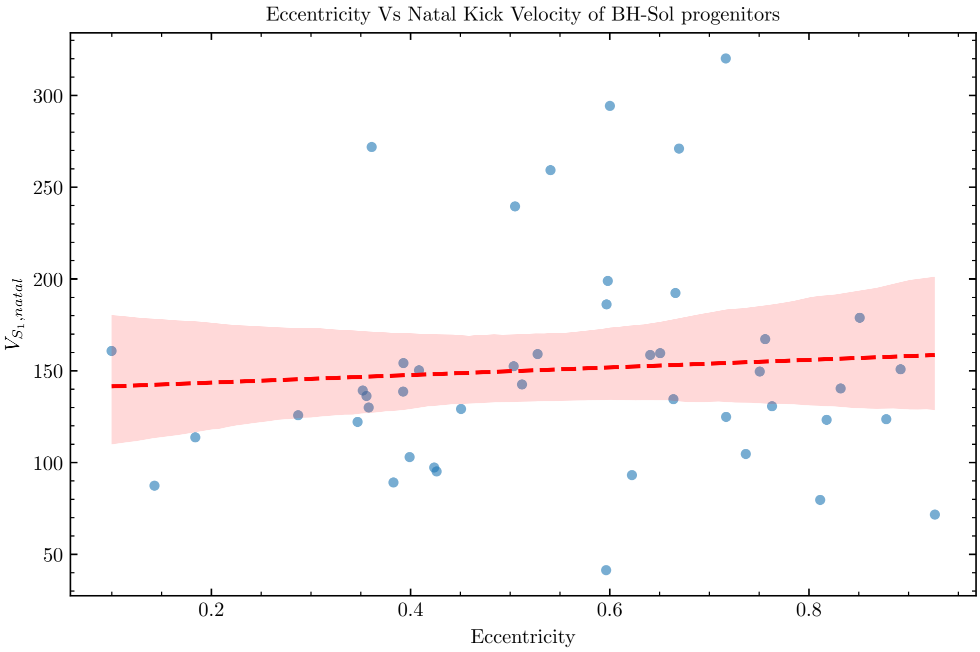
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